



Semiconductor Nanofab Training Simulator

Design Document



Design Concept

Introduction:

The Semiconductor Nanofab Training Simulator will be a digital, interactive guide aimed at more efficiently and adequately preparing and training students to use RIT's Semiconductor Nanofabrication Lab, alongside other potential use cases such as maintenance & navigation down the line. It will be available for download as a PC application, as well as potentially available on web through a browser.

Core Concept:

The simulator will have students navigating a digital twin of the Semiconductor Nanofabrication Lab (SNL), and replicating the real processes you might go through using the lab itself (gowning, operating a machine, disposing of garbage, ect) in a risk free digital environment. The processes will be explained, and students will be tasked to replicate them while making as few mistakes as possible.

Level Modules:

The training will be split up into "modules" (levels) that the user needs to go through and pass in order to prove that they understand the process at hand. Each module will teach the user how to perform a specific process, and the module itself will be split up into a number of "tasks," specific things the user has to do in-order before progressing. Once all tasks are complete, the user "passes" the module. These tasks are representative of definite steps pertaining to the process at hand, and each new step will be fully explained in-game when it is first introduced and as it is being performed for the first time.

Error Points:

Along with explanations, the training simulator works to teach users what not to do in the lab. When in a module, a value will tally any mistakes or "errors" the user has made, and if the user has messed up too much, they will have to restart the module from the start to pass. Each process will have a set of pre-determined errors associated with it, and each error will have a "severity" value associated with it based on how severe the mistake is. Every time an error is made, an "error score" tally will go up proportional to the severity of said error. If that tally exceeds an arbitrary threshold, the user will fail the module and be forced to restart if they want to complete it. If the module is passed, the simulator will store this information alongside the total elapsed completion time, and the best (lowest) passing error score as proof that the user understands and is adequately prepared to perform that process in the lab.

Intended Purpose & Structure

Learning Structure:

The intended design for the modules is that each module should build on the last, and what is explained in one will not be explained in the next. In other words, everything taught and graded on, including errors, will be explained once, but only once. The user will not be restricted in any way during any of the modules, but every single potential error or mistake will still be active in every single module, despite not being explained or taught in said module. For example, each module will start the user outside the lab, and the first module will teach them proper facility entrance and the gowning procedure, but the user will be expected to perform this gowning procedure without adequately failing at the start of every single module. If two steps are required to complete a certain task, and the user has already been taught the first, they will be expected to complete it without fail or explanation when being taught the second. This is by design, as it incentivises learning through repetition, and prevents users from just skimming through explanations or moving on from a module despite a shallow understanding, as they will be expected to adequately remember and perform it later on.

Module Progression:

There is a recommended order of progression through the modules for new users, however, all modules will be available for any user to try from the start, so that users more familiar with certain parts of the lab would be able to skip to more advanced modules without going through the previous ones, but users unfamiliar with the lab simply wouldn't know what to do in the later modules. There would still, of course, be a tutorial module on how to use the game's interface, as even if you were experienced with the actual Semiconductor Nanofab Lab, you might not know how to use or navigate the simulator game.

Mechanical Structure:

The SNL Training Simulator aims at being both as accessible as possible in regards to the technical and user experience, and as accurate as possible in simulating the real experience someone might have in the Lab itself. Balancing these two things is a tall task, as they often clash with one another, and the decisions made about the simulator are always informed by these two values. For example, the decision to design and build for PC and Web as opposed to something like Virtual Reality was made in favor of accessibility & ease of access for students with computers as opposed to the potential benefit of heightened accuracy and immersion offered by VR.

Game Mechanics

Menu & User interface:

Upon launching the application or opening on Web, the user will be presented with a main menu containing options to show available modules to start, resume a mid-progress paused module, read info about the lab itself, adjust settings, and quit. All modules will be accessible through this aforementioned list, which when clicked, shows the user a dropdown containing every single module separated into subcategories in their recommended order of completion. Each module item contains its name/title (pertaining to the process it aims to teach), its completion status (Completed, Incomplete or Previously Failed), the user's respective lowest error score (if applicable), the user's time of completion on that module. When clicked, the module-button sends the user into the respective module & the SNL Digital Twin itself. Accessible through a button on the side, each module in the list also contains a full transcribed explanation/instructions for the process at hand, as well real images of the associated machines, and possibly videos of the process being performed.

Interactable Objects:

When entering a level, the user will spawn outside the lab. You will play as a movable player character, around 6ft tall, and will have to look around (using the Mouse) and walk to (using WASD) the main door entrance. When looking at it, the door will highlight white, showing that you can interact with it. You do so by holding the E key, which highlights the door orange, showing that you are currently interacting with it, and allows you to move your mouse to the left or right to swing it open. Any interactable objects function like this. They will highlight white or orange to show that you can, or are currently interacting with them, whether that be picking up/holding an item, equipping a piece of clothing/equipment, or pressing a button on a machine.

Navigation:

Once your player enters the gowning room, you go through the proper gowning procedure before entering the lab and completing whatever task it is that you have to do next in your module. To navigate the facility, there will be temporary highlighted arrows on the floor leading the user to the location of the current task being explained. These arrows, however, will only appear when a task is explained for the first time, and there will be no map, as the user should learn to navigate the facility on their own.

Game Mechanics

Explanations:

When being led to do a task, there will be text on top of the screen with the name of the room and/or machine you have to go to telling you to go there, coinciding with the arrows. When at a machine, the text on top of the screen might prompt you to press a button, record a value, adjust a dial, or put an object in a specific place. Briefly explaining what you are doing, and highlighting in yellow the button, screen, dial, object, or place associated with the action you are doing to signify what you should do. This is pretty much how all explanations of specific tasks will function, text on top of the screen will explain to you what to do and why, which will coincide with some sort of visual guide like an arrow or a highlight. And when you complete said task, the text and visual guide will change, coinciding with the next task or step in that process.

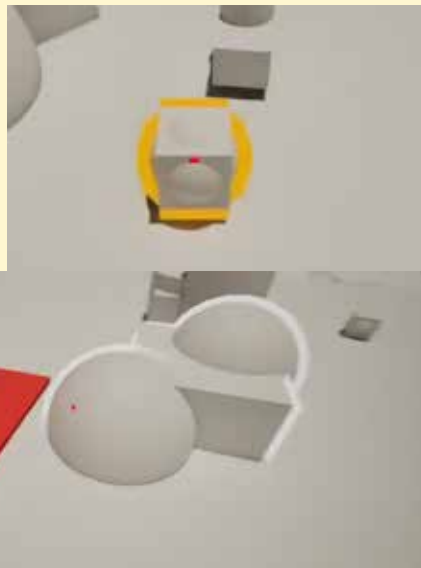
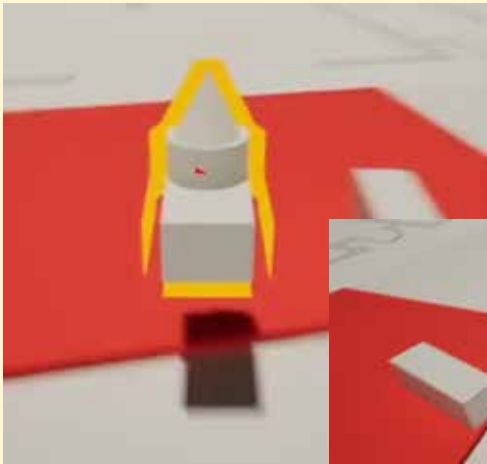
Task Structure:

There might be a handful larger overarching tasks per module, each larger task being explained with more detail beforehand like checking and adjusting a monitor to contextualize the smaller tasks like pressing a button and recording a value so that the user actually understands what they are doing in a broader sense instead of just mindlessly pressing buttons and knobs because the floating text tells them to. The user is still very hand-held throughout these explanations because in later modules they will need to remember and intuitively understand them without explanation.

Error and Task Failure:

Just like how when a task is completed there will be some sort of visual and auditory cue, the same is true for when an error occurs. The error tally will always be visible to the user, and when one occurs, whatever interaction or thing they did to cause the error will highlight in red, with text appearing on screen with the title of the error serving as a quick explanation, as well as brief text below it explaining what the error was in more detail, alongside the error severity/value added to the tally. And just like tasks, errors will always be explained before you have the chance to cause them. This does not mean errors will be inactive before they're explained, if you go around the lab pressing random buttons you will fail, it just means that errors pertaining to specific interactions and processes will be explained before you are specifically asked to perform those tasks. More specifically, the errors will usually be explained as you are performing those tasks. For example, if you are prompted to adjust a knob/setting or dispose of a liquid, you will also be prompted to not to adjust that knob too far, or dispose of that liquid in the trash, and exactly why.

Demo Images & Videos



Interactable
Mechanics Demo



Demo Walkthrough
of the Digital-Twin

